MILITARY SPECIFICATION

AMPLIFIER, AUDIO FREQUENCY, AM-3729( )/SR

1. SCOPE

1.1 This specification covers the design and performance of a 10 watt, solid state audio frequency amplifier for operation in conjunction with Navy communication receiving equipments having 600 ohm, audio frequency output impedances. It is intended for installation and use in Navy ships of all types and classes and shall operate, in accordance with this specification, under the adverse conditions normally encountered in a shipboard environment.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	—	Preservation, Methods of
MIL-S-901	—	Shock Tests, H.I. (High Impact) Shipboard Machinery, Equipment and Systems, Requirements for
MIL-D-3464	—	Desiccants; Activated, Bagged, Packaging Use and Static Dehumidifcation
MIL-E-16400	—	Electronic Equipment, Naval Ship and Shore: General Specification
MIL-E-17555	—	Electronic and Electrical Equipment; Accessories and Repair Parts, Packaging and Packing of
MIL-R-39008/2	—	Resistor, Fixed, Composition (Insulated), Established Reliability, Style RCR20
MIL-B-43291	—	Box, Fireboard, Corrugated, Double-Wall, Weather-Resistant.
MIL-I-45208	—	Inspection System Requirements

FEDERAL

PPP-B-601	—	Box, Wood, Cleated, Plywood
PPP-B-636	—	Box, Fiberboard
PPP-B-640	—	Box, Fiberboard, Corrugated, Triple Wall

STANDARDS

MIL-STD-105	—	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	—	Definitions of and Basic Requirements for, Enclosures for Electric and Electronic Equipment
MIL-STD-129	—	Marking for Shipment and Storage
MIL-STD-130	—	Identification Marking of U.S. Military Property
MIL-STD-147	—	Palletized and Containerized Unit Loads
MIL-STD-242	—	Electronic Equipment Parts (Selected Standards), Printed Wiring for Electrical Equipment
MIL-STD-461	—	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-781	—	Reliability Tests Exponential Distribution
MIL-STD-785	—	Reliability Program for Systems and Equipment, Development and Production

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PUBLICATIONS

HANDBOOKS

MIL-HDBK-217 — Reliability Stress and Failure Rate Data for Electronic Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, Room 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606).

3. REQUIREMENTS

3.1 General requirements. - The equipment shall be in accordance with the provisions of MIL-E-16400, except as hereinafter specified.

3.1.1 Material. - Material shall be in accordance with the requirements of MIL-E-16400 except as otherwise specified herein. Particular attention shall be paid to the acceptable and unacceptable materials of MIL-E-16400.

3.1.2 Selection of parts. - The selection and application of parts shall be in accordance with MIL-E-16400.

3.1.3 Temperature classifications. - The amplifier equipment shall conform with the requirements for class 4 of MIL-E-16400.

3.1.4 Test points. - Test points shall be provided in accordance with the requirements of MIL-E-16400.

3.1.5 Semiconductor devices. - The selection and application of all transistors and semiconductor devices shall be in accordance with MIL-E-16400. The use of electron tubes or nuvistors is not permitted under this specification.

3.1.6 Control knobs. - Control knobs shall conform with the requirements of MIL-E-16400. Tactile identification forms of control knobs will not be required.

3.1.7 Printed wiring. - Printed wiring, where employed, shall conform with the requirements of MIL-E-16400. Interconnection wiring between the multicontact mating connectors for the modular assemblies and the wiring to panel-mounted controls or to cabinet or chassis-mounted terminal strips shall be accomplished by conventional methods conforming to the requirements of MIL-E-16400.

3.1.8 Fuses. - Each side of the primary power input to the power supply shall be separately fused. The fuses shall be of the miniature, glass tube type and shall be mounted in insulated, extractor fuse holders mounted on the front operating panel.

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3.1.9 Blown fuse indicators. - Blown fuse indicators, selected from MIL-STD-242 shall be provided for each active fuse.

3.1.10 Time meters. - Elapsed time meters are not required.

3.1.11 Enclosure. - The degree of enclosure of the equipment housing shall be totally enclosed, in accordance with MIL-STD-108.

3.1.12 Securing devices. - All screws, bolts and nuts, (including front panel retaining devices) employed in the fabrication of the amplifier equipment, shall be of austenitic, corrosion resistant steel. Cadmium plated screws, bolts, nuts or other threaded hardware devices shall not be used, MIL-E-16400 or the Government Furnished Equipment notwithstanding.

3.1.12.1 All machine screws, other than front panel securing thumb screws, shall be of the panhead, crossed recessed drive, in accordance with MIL-E-16400.

3.1.12.2 Front panel retaining devices shall conform to the requirements of MIL-E-16400.

3.1.13 Resilient mountings. - The use of resilient mountings for equipment protection against the effects of shock and vibration are not permitted.

3.1.14 Finish. - All aluminum and aluminum alloy parts, both enameled and non-enameled, shall be chemically treated in accordance with the requirements of MIL-E-16400. External surfaces (cabinet and front panel) of aluminum and aluminum alloy parts shall be provided, in addition, with a primer and gray enamel coating in accordance with the requirements of MIL-E-16400.

3.1.15 Non-standard parts. - The preparation and submission of data necessary for the evaluation of non-standard electronic parts, shall be in strict accordance with the requirements of MIL-E-16400. Approval of non-standard parts, granted under prior procurements of the AM-3729( )/SR, or for non-standard parts employed in the Government Furnished Equipment, shall not be construed as being applicable to equipments covered by this specification, nor as waivers of any equipment thereof. Non-standard parts present in the Government Furnished Equipment shall not be employed in those applications where electrically and mechanically interchangeable standard parts are currently available.

3.1.16 Electrolytic capacitors. - The use of electrolytic capacitors of any description will not be permitted in any circuit directly identified with the basic audio amplifier. They may be employed in the power supply provided that their use does not compromise the requirements of this specification for reliability. Tantalum capacitors may be employed to the extent permitted by MIL-E-16400. The use of aluminum foil electrolytic capacitors will require specific approval in accordance with MIL-E-16400.

3.1.17 Special tools. - Securing devices which require special tools for tightening, loosening, or any other form of manipulation shall not be employed. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)). Standard tools, such as Allen or Bristol wrenches, shall not be furnished by the contractor.

3.2 Government Furnished Equipment (GFE). - Except as otherwise specified, herein, the audio frequency amplifier, covered by this specification, shall be equal to or better than the Government furnished equipment, wear, tear, damage and deterioration excepted.

3.2.1 A 15 ohm fixed resistor, $\frac{1}{2}$ watt, 5 percent tolerance, in accordance with MIL-R-39008/2 (FSN9N 5905-126-6691), circuit symbol R4, shall be inserted in series with pin 4 of connector J3, in equipments covered by this specification whether or not it is so employed in the Government furnished equipment. This component shall be installed externally with respect to any modular assembly.

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3.2.2 All parts, other than modular assemblies, shall be physically, mechanically and electrically interchangeable, part for part, with corresponding parts of the Government furnished equipment including extender cables.

3.2.3 All modular assemblies shall be physically, mechanically and electrically interchangeable, on a modular level only, with their counterparts of the Government furnished equipment. (See 3.12).

3.2.4 In the event of any conflict or inconsistency between the foregoing requirements for interchangeability and the performance and other requirements of this specification, the performance and other requirements of this specification take precedence and shall govern. The contractor shall determine whether any such conflicts or inconsistencies exist and shall notify the command or agency concerned giving full details, prior to the fabrication of the first article.

3.3 Equipment composition. - The audio frequency (AF) amplifier equipment shall be complete with all parts and items necessary for its performance in accordance with the requirements of this specification, when connected to the specified power source.

3.3.1 External cables for connection to the primary power source, to receivers, or ancillary equipment are not required to be furnished by the contractor.

3.3.2 Stuffing tubes are not required to be furnished by the contractor.

3.4 Overall dimensions and weight. -

3.4.1 Dimensions. - The maximum overall dimensions, exclusions, exclusive front panel handles (or other devices) shall not exceed the following:

Width	8 inches
Height	5 1/2 inches
Depth	5 7/8 inches

3.4.2 Overall Weight. - The maximum overall weight shall not exceed 12 pounds.

3.5 Primary power source. - The complete equipment shall be capable of operating from a primary power source of 105, 115, or 125 volts, 50 to 60 hertz or 400 hertz single phase, alternating current (ac) with no abrogation of performance by power line voltage variations of ±10 percent or power line frequency variations of ±5 percent, with either of these variations acting separately or in combination. Line voltage variations shall be considered with respect to the nominal voltage applied to the selected power transformer tap (see 3.5.2). Nominal line voltage is 115 volts. Nominal line frequency is 60 and 400 hertz.

3.5.1 The total power consumption at 115 volts, 60 or 400 hertz ac shall not exceed 30 watts.

3.5.2 The primary winding of the power transformer shall be electrostatically shielded from all secondary windings. The primary winding shall be provided with solder type terminals for each of the nominal primary power voltages specified in 3.5. Compliance with the performance requirements of this specification shall be predicated upon rated power source of 105 volts being applied to the 105-volt tap, 115 volts to the 115-volt tap, and 125 volts to the 125 volt tap. Selection of the appropriate primary tap shall be considered as an installation matter. Connection to the proper primary tap shall be effected by soldering methods. Mechanical links, switches or similar devices for changing taps are not required.

3.5.3 The power supply shall employ solid state devices for rectification and voltage or current regulation.

3.6 AF input. - The equipment shall incorporate facilities for the connection of two AF input lines to the input of the AF amplifier. Each line shall be considered as having a nominally rated impedance of

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600 ohms, at 1000 hertz and as being balanced with respect to ground. A selector switch, provided on the front operating panel, shall function to connect one of these two inputs to the input transformer of the AF amplifier. It shall function, also, to connect simultaneously a terminating load resistor of 600 ohms ±10 percent across the unused line input.

3.7 Audio input transformer. - The primary of the audio input transformer shall be electromagnetically balanced and center tapped, with the center tap brought out to a solder type terminal for grounding at the option of the installing activity. The primary of this transformer shall also be electrostatically shielded from the secondary winding.

3.8 AF output. - The output transformer of the AF amplifier shall be designed for operation into a balanced, 600 ohms ±10 percent, nominally rated load. The center tap of the secondary winding of the output transformer shall be arranged for optional grounding (see 3.8.1).

3.8.1 Audio output transformer. - The audio output transformer shall be provided with a secondary winding which shall be electrostatically shielded from the primary winding. It shall also be electromagnetically balanced with respect to ground, and its center tap shall be brought out to a separate solder type terminal for grounding, at the option of the installing activity.

3.9 Construction. - The design and construction of the amplifier equipment shall be predicated upon the use of modular assemblies of the "throw away" type, as employed in the Government furnished equipment.

3.9.1 Modular assembly functions. - The equipment shall contain three distinct modular assemblies (exclusive of front panel mounted controls and devices) which shall afford the following functional subdivisions:

(a) AF input, interstage amplification, and driver to power output stage
(b) AF power output stage
(c) Power supply

3.9.2 Equipment mounting. - The complete equipment shall be so designed and constructed that it may be mounted by means of bolts through the top, back or bottom surfaces to the cabinet assembly, to provide for bulkhead, overhead or top of table mounting, without the need for ancillary external or projecting mounting brackets, and without the need for resilient mounts for protection against shock or vibration.

3.9.2.1 The design and construction of the complete equipment shall be such that when the equipment is mounted, in any manner described in 3.9.2, it shall be capable of withstanding exposure to shock and vibration, as specified herein, without damage and without evidence of cabinet distortion, tearing of metal surrounding the mounting bolts, damage to or loosening of any part, or any compromise of the free closing, opening, latching, or withdrawal of the chassis panel assembly.

3.9.3 Cabinet assembly. - The cabinet assembly shall be a five-sided enclosure, and shall be essentially of one-piece construction. This assembly shall house the chassis panel assembly which shall form the sixth side of the enclosure when the chassis-panel assembly is secured in its closed position.

3.9.3.1 The cabinet assembly and all items entering into its construction shall be fabricated from aluminum alloy. Hinges, latches, and other hardware items are not included in this requirement.

3.9.3.2 The cabinet assembly shall be of sturdy design and construction, compatible with the type of service for which it is intended and the requirements of this specification for equipment mounting.

3.9.3.2.1 Seams of cabinet fabricated from sheet metal shall be flame welded, or similarly secured to assure positive electrical contact throughout their length. If spot welding is employed, such welding shall be accomplished by spacing the welds at intervals not greater than 3/4 inch.

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3.9.3.3 The front opening of the cabinet assembly shall be of such design as to provide positive electrical bonding between the basic cabinet assembly and the front chassis panel assembly. The mating of these two assemblies when in the operating condition, shall present flush surfaces between the sides, top, and bottom of the corresponding mating edges of the two assemblies involved.

3.9.4 Chassis panel assembly. - The chassis panel assembly, with its contained parts mounted thereon, shall be so designed and constructed as to become an integral part of the cabinet assembly. It shall be so attached to the basic cabinet assembly as to permit ready and easy removal without disturbing the installation of the latter.

3.9.4.1 The chassis-panel assembly shall be of unit construction, and shall serve not only as an operating panel but also as the basic chassis structure for the containment of the modular assemblies required for the complete equipment.

3.9.4.2 The chassis-panel assembly and all brackets, shields, and cover platen shall be fabricated from aluminum alloy.

3.9.4.3 The front operating panel shall be permanently secured to the chassis, and shall be designed and constructed to engage the front opening of the basic cabinet structure in a manner which affords positive electrical bonding between the panel and this latter structure.

3.9.4.4 The front panel shall be secured to the cabinet structure by means of a hinge, at the base of the cabinet (lengthwise), and two captive, knurled and slotted head screws at the upper corners of the panel. These screws shall be captive with the front operating panel, and shall engage threaded inserts seated in flanges or as otherwise provided in the basic cabinet structure.

3.9.4.5 The base hinge or hinges, (if furnished as separate hardware) shall be corrosion-resistant steel with removable corrosion-resistant steel pins, which can be readily removed without dependence upon open clearances beyond the base dimensions of the cabinet assembly. The hinge (or hinges) may be integral with the front panel and the cabinet assembly. In this latter event, the retaining pins shall be of corrosion-resistant steel.

3.9.4.6 Sliding, locking type latches shall be provided to secure and to support the chassis panel assembly on its opened position. The latches shall permit the chassis panel assembly to swing downward, through a full 90 degrees with respect to its base-mounted hinge or hinges. The latches shall be of such design and construction that they may be readily detached from the cabinet assembly to permit the removal of the chassis-panel; to lock positively the chassis-panel assembly in its opened, horizontal position; and to preclude any damage to the chassis-panel assembly or any damage or loosening of any of its contained parts upon exposure of the equipment to shock or vibration, as specified herein, when the chassis-panel assembly is locked in its opened position. The latching devices shall be fabricated from corrosion-resistant steel. The foregoing requirement for a full 90-degree downward swing of the chassis-panel assembly shall not be construed to mean that the front panel mounted controls and devices need to be recessed.

3.9.4.7 The front panel shall be provided with a pull handle to facilitate the opening and closing of the chassis-panel assembly without subjecting the control knobs or other panel-mounted devices to undue strain. The pull handle shall be capable of supporting the full weight of the complete equipment.

3.9.4.8 The design of the panel-chassis assembly shall be such that it may be placed on a flat surface, on its front, rear, or either side without danger of damage to or derangement of any of its parts or assemblies.

3.9.4.9 The modular assemblies shall be integral elements of the chassis-panel assembly. Each modular assembly shall be accessible for ready removal and replacement from the front of the equipment

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when the chassis panel assembly is secured in its opened position. Each modular assembly shall be individually removable or replaceable, without disturbing any other modular assembly, upon the loosening of its retaining devices, and shall not require the use of a soldering iron or the disconnection of external leads or wiring.

3.9.5 Electrical interconnections. -

3.9.5.1 Interconnection of modular assemblies. - Electrical interconnections between the modular assemblies panel-mounted controls and devices, and external cable terminal strips shall be effected through the use of self-aligning, multi-contact connectors. The modular assemblies shall be fitted with male type, multi-contact connectors for mating with corresponding female type multi-contact connectors mounted on the chassis structure. The female type multi-contact connectors shall contain floating type sockets. The connectors, both male and female, shall conform to the requirements of MIL-E-16400 with respect to their individual design, construction, materials, keying, and protective sleeves for the preclusion of damage from malinsertions. Mating pins and sockets in the connectors shall be gold plated.

3.9.5.2 Electrical connectors. - External cables (power input, AF input and AF outputs) shall terminate at a suitable terminal strip located at the rear of the cabinet assembly. The terminal strip shall contain solder type terminals for these cable terminations. The terminal strip shall be mounted on supports so as to project no less than $\frac{1}{2}$ inch into the enclosure from the rear mounting surface. The terminal strip shall comply with the requirements of MIL-E-16400 with respect to materials, markings, and provision for spare terminals. The terminal strip shall be positioned so that its solder type terminals are accessible for lead connection and soldering from the front opening of the cabinet assembly, when this latter assembly is in its installed position.

3.9.5.3 Chassis interconnection - Interconnection between the chassis-panel assembly and the rear mounted terminal strip (see 3.9.5.2) shall be effected by means of a multi-conductor cable, one end of which shall be permanently connected to the terminal strip, with the other end being terminated at the chassis panel assembly through multicontact plug and jack connector. The cable plug shall be of the male type and the mating female type connector shall be mounted on the chassis panel assembly. The male and female connectors shall conform to the requirements of 3.9.5.1.

3.9.5.3.1 The interconnecting cable shall be so positioned and be of such length and flexibility so as not to restrict the free opening and swing of the chassis panel assembly; and to preclude damage to itself from the opening or closing of the chassis panel assembly.

3.10 Panel controls and devices. - The front operating panel shall contain the controls and devices specified hereinafter. Locking devices for controls other than keying, are not required.

3.10.1 On-off switch. - The on-off switch shall be a toggle switch operating to apply or remove primary power. It shall be entitled POWER with the two positions of the toggle marked ON and OFF with ON applying to the up position of the switch toggle.

3.10.2 AF level control. - The AF level control shall be a potentiometer, entitled "AF Level" and shall be provided with a scale consisting of eleven major divisions and ten alternate minor divisions with the major divisions marked from 0 to 10, inclusive, in clockwise sequence.

3.10.3 Channel selector switch. - The channel selector switch shall be a two-position, rotary type switch operating to permit the selection of either one of the two AF input channels. The control shall be entitled "Channel Selector" with its two positions marked 1 and 2, in clockwise sequence.

3.10.4 Indicator lamp. - A 6 to 8 volt, bayonet base lamp contained within an approved holder fitted with a red lens and mechanical shutter shall be furnished. The lamp shall be replaceable from the front of the equipment without opening the chassis panel assembly from the cabinet assembly. The lamp holder shall be in accordance with MIL-E-16400. The indicator lamp shall have no title or markings.

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3.10.5 Fuses. - The two active and spare fuses mounted on the front panel (see 3.1.8) shall have no title. The spare fuseholder shall be marked "Spare".

3.11 Terminal tube entrances. - The design of the cabinet assembly shall be such that holes for terminal tube entrances may be drilled by the installation activity through either of the two sides, the top, the bottom, or the rear of the cabinet.

3.12 Soldering. - Soldering of wiring to terminals, connectors, and so forth, shall conform with the requirements of MIL-E-16400. Insulating sleeves shall be installed in all pins and jacks and all other closely spared solder type terminations.

3.13 Approval of modular assemblies, packaging and interconnecting techniques. - The modular assemblies, packaging and interconnecting techniques shall be submitted to the command or agency concerned for approval in accordance with the requirements of MIL-E-16400.

3.13.1 The interchangeability requirements of 3.2 dictate sealed modular assemblies. The component parts of these assemblies, if non-standard, need not be submitted for approval, under the requirements of MIL-E-16400 for non-standard parts approval. In lieu, thereof, approval for these parts will be contingent upon the requirements of MIL-E-16400 for modular assemblies approvals. (See 3.13).

3.14 Performance requirements. - The equipment shall comply with the performance requirements specified under the standard test conditions (see 4.1.4) unless otherwise specifically stated or required for performance evaluation.

3.14.1 The circuitry provided in the equipment shall incorporate electrical elements, including power supply regulation, automatic adjustment of gain as appropriate, temperature compensation, as needed, and choice of circuit parameters and constants which will assure the conformity of the equipment with the performance requirements of this specification. These requirements shall not be abrogated by the use of parts, including transistors and diodes, having electrical tolerances which are within the limits specified in their applicable part specifications.

3.14.2 Input impedance. - The input impedance of each AF input shall be 600 ohms ±10 percent as measured at 1000 hertz. This impedance shall prevail regardless of the adjustment of the AF level control.

3.14.3 Channel selector switch characteristics. - The channel selector switch shall conform to the requirements of MIL-E-16400. It shall be so arranged as to provide a nominal 600 ohms ±10 percent noninductive resistive load, as measured at 1000 hertz for the AF channel not connected to the input transformer of the AF amplifier. Both sides of each AF input channel shall be considered to be above ground potential. It is not required that the noninductive resistive load be center tapped.

3.14.4 Audio power input. - The AF amplifier shall be capable of producing rated power output with any adjustment of audio power input from 0.006 to 2 watts, at 1000 hertz. A corresponding adjustment of the AF level control to accommodate this range of AF input variation is permitted.

3.14.4.1 Automatic regulation of the gain of the input AF amplifier to permit full range of adjustment of the AF output level for any level of AF input without overload of the input AF amplifier stage shall be provided.

3.14.5 AF output impedance. - The secondary winding of the AF output transformer shall have a nominal impedance of 600 ohms ±10 percent as measured at 1000 hertz.

3.14.6 Audio power output. - The amplifier shall be capable of providing rated power output, as measured at 1000 hertz across a nominal noninductive resistive load of 600 ohms ±10 percent. Rated power output shall be not less than 10 watts.

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3.14.7 Distortion. - The overall distortion of the amplifier equipment, as determined from a measurement at the AF output terminals and under conditions of adjustment of the amplifier for rated power output, shall not exceed 2 percent, root mean square (rms) value, for any input level within the limits specified in 3.14.4. This requirement shall be met for audio frequencies of 200, 1000, and 4000 hertz. Allowance is permitted for changes in output level due to audio attenuation at 200 and 4000 hertz.

3.14.8 Frequency response. - The overall frequency response of the amplifier from either of the AF input channels to the output terminals shall be flat, within ±2 decibels (dB) between the limits of 200 and 4000 hertz. Rated AF power output at 1000 hertz shall be taken as the zero decibel reference level in determining compliance with this requirement. The frequency response below 200 hertz down to 30 hertz and above 4000 hertz up to 10,000 hertz shall demonstrate increasing attenuations. The specified limits for attenuation (in decibels) are with respect to voltage ratios.

3.14.9 Signal plus noise-to-noise ratio. - The signal plus noise-to-noise ratio (in volts) shall not be less than 60 dB as measured for rated AF power output and for an AF input of 6 milliwatt (mW).

3.14.10 Output regulation. - The AF amplifier shall incorporate output regulation operating to maintain the output voltage constant within 2 dB for a change in output load from 600 to 300 ohms.

3.14.11 Microphonics. - There shall be no evidence of microphonics with the amplifier operated under conditions of full maximum audio output with the AF level set at a maximum level position or upon subjection of the equipment to shock and vibration, as specified herein.

3.14.12 AF level control characteristics. - The AF level control shall operate to provide an attenuation range of not less than 80 dB in audio power output. Not less than 70 percent of the angular rotation of the control shall be required to reduce the audio output power 40 dB below rated AF output power. These requirements shall be met with the amplifier initially adjusted to produce rated power output across a 600 ohm noninductive resistive load.

3.14.12.1 The AF level control shall conform with the following requirements for linearity.

(a) A curve plotted on linear graph paper of AF output power level, in decibels, versus percent of angular rotation of the control shall deviate less than 6 dB from a straight line drawn through the point on the actual curve which represents the departure from rated AF power output and the point on the curve which represents a 40 dB attenuation below such output power.
(b) The actual plot shall be a smooth curve between the points representing 40 dB and 80 dB attenuation from rated AF power output. No limits for linearity is specified for the region of the curve between the 80 dB and 40 dB points of attenuation.

3.14.13 Line voltage variations. - The requirements of 3.14.2 thru 3.14.11 shall not be abrogated when the equipment is subjected to the combinations of changes in line voltage and frequency or the application of transient voltage or frequencies, as specified in MIL-E-16400. This requirement is applicable to the following nominal (or rated) conditions as reference points.

(a) 115 volts, at 60 hertz applied at the 115 volt tap on the power transformer
(b) 115 volts, at 400 hertz, applied at the 115 volt tap on the power transformer

3.14.14 Variation in ambient temperature. - There shall be no evidence of overheating or degradation of the performance requirements of 3.14.2 thru 3.14.11 when the equipment is subjected to variations in ambient temperature, as specified in MIL-E-16400.

3.14.15 Variations in humidity. - There shall be no evidence of rust, corrosion or degradation of the performance requirements of 3.14.2 thru 3.14.11 when the equipment is subjected to variations in relative humidity, as specified in MIL-E-16400.

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3.14.16 Inclination, shock and vibration. - The equipment shall be so constructed as to be capable of rough handling during shipment, installation and operation. It shall be capable of meeting the performance requirements of this specification after exposure to the following:

(a) Inclination as per MIL-E-16400
(b) Shock as per MIL-E-16400 and MIL-S-901 for Grade A, hull mounted, Class I, Type A, Lightweight Equipment Shock
(c) Vibration as per MIL-E-16400

3.14.17 Reliability. - The requirements specified herein shall take precedence and shall govern over the requirements of 3.2 thru 3.2.2.

3.14.17.1 Quantitative reliability requirement. - The equipment shall have a specified (θ_0 as defined in MIL-STD-781) mean-time-between failure (MTBF) of 3000 hours.

3.14.17.2 Reliability program plan. - The manufacturer shall establish and implement a reliability program plan in accordance with MIL-STD-785. A written description of the program plan shall be submitted to the command or agency concerned.

Particular attention shall be paid to those sections of MIL-STD-785 which require submission of detailed tasks, procedures to implement and control these tasks and time schedules. The reliability program plan, as approved, becomes the basis for contractual compliance. The plan, including reliability test plans, shall describe an integrated effort within the total project; it shall provide information to show how the manufacturer shall meet specified quantitative reliability requirements during production.

3.14.17.3 Reliability prediction. - The manufacturer shall submit a design reliability prediction, in accordance with the procedure described in MIL-HDBK-217 based on stress factors and part populations, to the command or agency concerned for approval. The prediction shall assume a 65°C ambient temperature and a shipboard environment. The processes and procedures required to assure production of reliable equipment shall be thoroughly documented.

3.14.17.4 Source of failure rates. - The primary source of failure rate data shall be MIL-HDBK-217. For items not included in MIL-HDBK-217, other sources of data may be used with the approval of the command or agency concerned. When such alternate sources of data are proposed, enough information shall be submitted to enable the command or agency concerned to evaluate the validity and applicability of the results derived therefrom.

3.14.17.5 Reliability reports. - Reports providing complete accounting of progress on each element defined by the reliability program, results achieved, and status of action to resolve problems and correct weak links shall be submitted monthly by the command or agency concerned. The results of the demonstration test shall be reported in accordance with MIL-STD-781.

3.14.18 Burn-in operation. - Each equipment shall be capable of a pretest "burn-in" consisting of a continuous period of seventy-two (72) hours of operation at 10 watts AF output without evidence of overheating or loss of output level in excess of 0.5 dB of the value which prevailed prior to the start of the burn-in period without adjustment of the audio level control.

3.15 Magnetic characteristics. - No additional quantities of magnetic materials shall be added to design over and above that now in the Government furnished equipment or as otherwise specified there in.

3.16 Identification of equipment. - The equipment shall be marked for identification in accordance with MIL-STD-130.

3.17 Safety. - The design, development and production shall promote maximum safety of both

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operational and maintenance personnel and equipments during all phases of operational life in accordance with MIL-E-16400.

3.18 Workmanship. - Workmanship shall conform with the requirements of MIL-E-16400.

3.19 Synchro transmitters and receivers. - Synchro transmitters and receivers are not required.

3.20 Motors and generators. - Motors and generators are not applicable.

3.21 Universal motors, dynamotors, synchronous converters, and similar rotating electrical devices. - Universal motors, dynamotors, synchronous converters, and similar rotating electrical devices are not applicable.

4. QUALTIY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractors quality assurance. - The contractor shall provide and maintain an effective inspection and quality assurance system acceptable to the Government. The inspection system shall conform to the requirements of MIL-I-45208. A current written description of the means by which the contractor will assure the quality of products shall be submitted to the command or agency concerned for evaluation. The written description shall include flow charts showing complete flow of material from inspection on receipt through all manufacturing processes to final shipment. The various process and inspection positions shall be clearly identified on the flow charts and cross-referenced to the detailed inspection procedures and criteria of the written description. Any changes to the procedures or the described system shall be submitted to the command or agency concerned for evaluation prior to use.

4.1.2 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practice, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all the requirements of this specification. Failure of the contractor to correct promptly, deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Test procedures. - The contractor shall submit to the procuring activity, via the Government inspector for review and approval, written test procedures including test equipment and tolerance limits to be used for first article inspection and quality conformance inspection. Each test shall be identified in accordance with paragraph 4.2 as to classification.

4.1.3.1 Procedures for inspection, tests, demonstrations, and so forth, shall contain step by step information which will enable the operator to properly perform the requirements of this specification. The method of examination shall be clearly defined.

4.1.3.2 The procedures shall contain a list of test equipment identified by name and model. Special test equipment shall also be listed and identified.

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4.1.3.3 Data sheets shall be included in the procedures. Conformance parameters shall be clearly indicated with tolerance limits. Whenever quantitative measurements are made, the result shall be shown by quantitative data.

4.1.4 Standard test conditions. - Unless otherwise specified, test measurements shall be made under the following conditions.

(a) Power supply input - 115 volts, 60 hertz, single phase, ac
(b) Input impedance - 600 ohms, ±10 percent, noninductive resistive load
(c) Output impedance - 600 ohms; ±10 percent, noninductive resistive load
(d) Ambient temperature - between 20 and 30°C
(e) Relative humidity - prevailing

4.2 Classification of inspection. - The method of examination and testing of the equipment fall within the following classification:

(a) First Article Inspection
(b) Quality Conformance Inspection
    (1) Production Inspection
    (2) Production Control Inspection
    (3) Environmental Inspection
    (4) Inspection of Preparation for Delivery
(c) Reliability Testing

4.3 First article inspection. - First Article Inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First Article Inspection shall include the tests and examinations specified in Table I:

Table I - First Article Inspection

Inspection requirement	Reference
Production Inspection	4.4.1
Production Control Inspection	4.4.2
Environmental Inspection	4.4.3
Accelerated life test	MIL-E-16400
Overall dimensions and weight	3.4.1 and 3.4.2
Safety	3.17

4.3.1 First article inspection report. - A first article inspection report shall be prepared containing all the results of the first article inspection. The report shall include completed Form NAVEXOS 3373 as specified in MIL-S-901 for the shock test. The first article inspection report shall be submitted to the procuring activity, via the Government inspector, for review and approval. The first article inspection report shall indicate specification compliance limits as well as actual performance test results. The contractor shall certify that, on the basis of the report, the equipment does or does not comply with specification requirements.

4.4 Quality conformance inspection. -

4.4.1 Production inspection. - Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which

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detect hidden defects of material. Production inspection shall consist of the tests approved in accordance with paragraph 4.1.3 and shall include the requirements of Table II. Production Inspection shall be performed after the burn in operation of paragraph 4.4.1.1 of this specification.

Table II - Production Inspection

Requirement	Reference
(a) Audio power output	3.14.4 and 3.1.4.6
(b) Distortion	3.14.7
(c) Frequency response	4.4.1.4
(d) Signal plus noise to noise ratio	3.14.9
(e) Output regulation	3.14.10
(f) Qualitative operation of blown fuse indicators	4.4.1.2
(g) Operating test	4.4.1.3
(h) Each equipment shall be examined for the following:	
(1) Workmanship in accordance with MIL-E-16400 (see 3.18)	
(2) Assembly; enclosure (see 3.1.11) and fit	
(3) Marking (see 3.16)	
(4) Materials; parts and finish (see 3.1.14)	
(5) Treatment for prevention of corrosion	
(6) Soldering (see 3.12)	

4.4.1.1 Burn in operation. - Each equipment shall be subjected to a burn in as specified in 3.14.18. The power output measurement shall be made with an input signal of 6 milliwatts at 1000 Hz. Measurements shall be made at least once during each 24 hour period with the last measurement occurring just prior to completing the 72 hour period. The burn in operation may be discontinued over a weekend.

4.4.1.1.1 Failure. - A burn in operation failure is a loss in excess of 0.5dB of the output power (see 3.14.18). Should a failure occur, it shall be the contractors option to:

(a) Repair the failed part and resume test within a two hour period.
(b) Replace a failed module with one having an equivalent number of accrued test hours.
(c) Replace a failed module with a newly fabricated module and forfeit the accrued hours on the total equipment under test by restarting the test with zero hours accrued.

4.4.1.2 Qualitative operation of blown fuse indicators. - A test shall be performed that will qualitatively show correct operation of the blown fuse indicator.

4.4.1.3 Operating test. - The equipment shall be assembled in the end item configuration required for delivery and subjected to an operating test to insure qualitatively the proper functioning of the equipment, including all operating controls and cables.

4.4.1.4 Frequency response. - The overall frequency response of the amplifier shall be measured at 200, 500, 1000, 2000 and 4000 Hertz (see 3.14.8).

4.4.2 Production control inspection. - Production control inspection, including sampling, shall conform to Table III below and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. Production control inspection shall be approved in accordance with 4.1.3. The equipment shall satisfactorily meet the requirements of Production Control Inspection prior to release for shipment.

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Table III - Production Control Inspection

Inspection	Inspection Paragraph	AQL—%
(a) Supply line voltage and frequency	4.4.2.7	6.5
(b) Heat test	4.4.2.3	6.5
(c) Power test	3.5.1	6.5
(d) Interchangeability	4.4.2.5	6.5
(e) Microphonics	4.4.2.4	6.5 for the group
(f) Input impedance	3.14.2	
(g) Audio output impedance	3.14.5	
(h) Level control characteristics (attenuation and angular rotation)	3.14.12	
(i) Channel selector switch characteristics	3.14.3	
(j) Frequency response	4.4.2.6	

4.4.2.1 Rejected lots. - If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.2.2 Reinspection of conforming production control test units. - Unless otherwise specified, sample units which have been subjected to and passed production control inspections may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all damage.

4.4.2.3 Heat test. - The heat test shall be performed in accordance with MIL-E-16400. The equipment shall operate under conditions of 6 milliwatts, 1000 Hertz input with the output adjusted for 10 watts. The following requirements shall be met at the end of the test.

(a) Audio power output (see 3.14.6)
(b) Distortion (see 3.14.7)
(c) Frequency response (see 3.14.8)
(d) Signal plus noise to noise ratio (see 3.14.9)
(e) Visual examination to determine that damage or deterioration did not occur

4.4.2.4 Microphonics. - A test shall be performed to determine the absence of microphonics (see 3.14.11).

4.4.2.5 Interchangeability. - Modular assemblies produced and modular assemblies from the government furnished equipment shall be interchanged to determine physical, mechanical, and electrical interchangeability. The electrical tests specified in 4.4.2.3 shall be performed to determine electrical interchangeability (see 3.2 and 3.9.1). Interchangeability shall be demonstrated on the first production lot and on every other subsequent lot.

4.4.2.6 Frequency response.- The frequency response from 200 down to 30 Hertz and from 4000 to 10,000 Hertz shall show increasing attenuation (see 3.14.8). Reference data shall be taken at 200 and 4000 Hertz. Measurements shall be made at the following frequencies:

Low end (Hertz)	High end (Hertz)
200	4000
100	7000
50	10000
30	

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4.4.2.7 Supply line voltage and frequency. - The equipment shall be subjected to tests to determine conformance to the requirements of 3.14.13. The equipment shall be initially conditioned for 10 watts output at 1000 Hertz with 6 milliwatts input at nominal line voltage and frequency. The AF level control shall not be readjusted during test. The specified requirements shall not have degraded more than indicated from the measured values at nominal voltage and frequency when measured as specified in 3.14.13b.

(a) AF power output (see 3.14.6) 0.5dB
(b) Distortion (see 3.14.7) ±0.5 percent
(c) Signal plus noise to noise ratio (see 3.14.9) ±5dB

4.4.3 Environmental inspection. - Environmental inspection shall be as listed in Table IV and shall be performed on units which have been subjected to and passed production inspection. Environmental tests shall be approved in accordance with 4.1.3.

Table IV

Group A Environmental Tests

Inspection	Reference
Temperature	4.4.3.4
Vibration	4.4.3.5
Inclination	4.4.3.6
Magnetic test	MIL-E-16400

Group B Environmental Tests

Inspection	Reference
Humidity	4.4.3.8
Shock	4.4.3.7
Salt spray	MIL-E-16400
Weld test	MIL-E-16400

4.4.3.1 Sampling for environmental inspection. -

4.4.3.1.1 Group A environmental tests. - One equipment shall be randomly selected from each successive 150 units produced or fraction thereof. The first sample shall be selected from the first month's production.

4.4.3.1.2 Group B environmental tests. - One equipment shall be randomly selected from each successive 400 units produced or fraction thereof. The first sample shall be selected from the first month's production.

4.4.3.2 Nonconforming environmental sample units. - If a sample unit fails any test specified in paragraph 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see paragraph 6.3).

4.4.3.3 Reinspection of conforming environmental sample units. - Unless otherwise specified in the contract, sample units which have been subjected to and passed environmental tests may be accepted on the contract provided they are resubjected to and pass production inspection after thorough reconditioning.

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4.4.3.4 Temperature. - The equipment shall be subjected to tests to determine conformance to the temperature requirements of 3.14.14 (see 3.1.3). No readjustment of the level control is permitted after the test has begun. After completion of the temperature test, the equipment shall be subjected to a visual examination to determine any damage or deterioration.

4.4.3.5 Vibration. - The equipment shall be subjected to tests to determine conformance to the vibration requirements of 3.14.16. After completion of vibration, the equipment shall be examined for conformance with 3.9.2.1, 3.9.4.6, and 3.14.11.

4.4.3.6 Inclination. - The equipment shall be subjected to tests to determine conformance to the inclination requirements of 3.14.16. Electrical tests as specified in 4.4.2.3 shall be met throughout the test.

. . 4.4.3.7 Shock. - The equipment shall be subjected to tests to determine conformance to the shock requirements of 3.14.16. After completion of the shock test the equipment shall be examined for conformance with 3.9.2.1, 3.9.4.6, and 3.14.11. Equipment shall meet the test requirements of 4.4.2.3 after shock.

4.4.3.8 Humidity. - The equipment shall be subjected to tests to determine conformance to the humidity requirements of 3.14.15. No readjustment of the level control is permitted after the test has begun. After completion of the humidity test the equipment shall be examined for conformance with 3.14.15.

4.4.4 Inspection of preparation for delivery. -

4.4.4.1 Prior to packaging the equipment, examination shall be made to determine that the equipment is in the following condition for shipment:

(a) Power transformer. The power transformer shall be electrically connected for an input voltage of 115 volts, ac (see 3.5.2).

4.4.4.2 Examinations shall be conducted to insure conformance with the requirements of section 5 of this specification.

4.5 Reliability testing. -

4.5.1 Reliability test plans. - The contractor shall submit to the procuring activity a reliability demonstration test plan which conforms to the detailed requirements of MIL-STD-781 and the requirements of this specification.

4.5.2 Qualification test. - The qualification test shall be conducted on the first 5 production units. The test will be of the fixed time type. The test shall run for 2300 unit hours with zero failures to demonstrate 1000 hours MTBF to a 90% lower confidence level. Input voltage cycling, as specified in MIL-STD-781, is required.

4.5.3 Sampling phase of reliability tests. - Five units shall be randomly selected by the Government inspector from each 1000 units produced for a reliability test to be conducted as specified (see 4.5.1 and 4.5.2).

4.5.4 Failure. - Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth in this specification.

4.5.5 Rejected equipment. - When equipment has been rejected as a result of the reliability test, the whole lot from which it was drawn shall be rejected. In the event of rejection, the lot of equipments shall be reworked as necessary, reinspected, and another random sample drawn for another reliability

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test. No equipment shall be selected more than once for reliability test. In the event that the cause of failure is equipment manufacturer design or part manufacturer design, corrective action shall be performed on all equipments delivered under the contract. If failure is due to equipment manufacturer workmanship or part manufacturer workmanship, the entire inspection lot shall be inspected for the discovered fault and reworked as necessary.

4.5.6 Disposition of tested equipment. - Upon successful completion of reliability testing, tested equipment shall be returned to the production area for reinspection and reconditioning, as required, and shall be subjected to the production tests of 4.4.1 to assure that it meets the requirements of this specification. After reconditioning, reinspection, and retesting the equipment may be delivered as part of the contract quantity provided it meets all terms and conditions of the contract for acceptance.

4.5.7 Test Parameters. - As a minimum, the parameters to be monitored during reliability testing shall be as follows:

Inspection	Reference Paragraphs
AF Power Output	3.14.6
Distortion	3.14.7
Signal Plus Noise-to-Noise Ratio	3.14.9

4.6 Shock-tested sample equipments shall be thoroughly reconditioned by the contractor before submission as production equipments.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. - Each set shall be individually packaged in accordance with Method 11b of MIL-P-116. Desiccant shall conform to MIL-D-3464, Type II or III. Loose-fill polystyrene shall not be used for packaging. The container exterior to the barrier shall conform to V-grade of PPP-B-636, or MIL-B-43291.

5.1.1.1 Technical literature. - Technical literature shall be packed in transparent barrier material minimum 4 mil thick. Closure shall be by heat sealing. The technical literature package shall not be placed within the barrier material used to package any other items.

5.1.1.2 Equipment repair parts. - Equipment repair parts or running spares shall be individually packaged in sealed transparent packages in accordance with MIL-E-17555. These items shall be packaged separately from other items within the pack in containers conforming to PPP-B-636, class weather resistant and identified to the applicable equipment. Repair parts lists shall be provided in accordance with MIL-E-17555.

5.1.2 Level C. - Packaging shall be sufficient to afford adequate protection against corrosion, deterioration and any physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. - Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. - Items packaged as specified in 5.1 shall be packed in containers conforming to PPP-B-601, Overseas Type, except that cleats will not be used on top panels. Strapping shall be in accordance with the appendix of PPP-B-601.

5.2.2 Level B. - Items packaged as specified in 5.1 shall be packed in containers conforming to MIL-B-43291 or PPP-B-640, class 2. Closure and reinforecement shall be in accordance with the applicable box specifications, except that only non-metallic tape banding shall be used for reinforcement.

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5.2.3 Level C. - Items packaged as specified in 5.1 shall be packed in containers in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.2.4 Unitized loads. - When applicable, items packed as specified in 5.2 shall be palletized or containerized in accordance with MIL-STD-147, whenever quantities to one destination equal 50 cubic feet or more. Quantities of packed items too small or impracticable for a unitized load shall be consolidated in shipping containers specified in 5.2. Palletized loads shall be strapped to the pallet. Unitized loads shall not be subjected to rough handling tests. When applicable, items for shipboard end destinations shall be unitized by hull number.

5.3 Marking. - In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129, except that all exterior Level A and B pack markings shall be waterproof.

5.4 Inspection. - Inspection of Level A packages shall be in accordance with MIL-P-116.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

6.1.1 Procurement requirements. -

(a) Title, number, and date of this specification
(b) Selection of applicable level of packaging, packing or marking requirements other than those required by section 5 (see 5.1 and 5.2).

6.1.2 Contract-data requirements. - Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423). The data required by this specification is as follows:

(a) Reliability program plan (see 3.14.17.2)
(b) Reliability prediction (see 3.14.17.3)
(c) Test procedures (see 4.1.3)
(d) Reliability test plan (see 4.5.1)

6.2 Definitions. -

6.2.1 Rated power output. - The rated power output is the root mean square (rms) power output (including distortion) which the amplifier will deliver, as rated distortion, into its rated noninductive load impedance when the amplifier is operated from its rated primary power source.

6.2.2 Distortion. - Distortion is the difference between the harmonic content of the voltage at the output terminals and that of the input terminals and is expressed as a percentage of the rms output voltage.

6.2.3 Frequency response. - The frequency response of the amplifier is the variation of gain as a function of frequency over the frequency range specified herein. It shall be expressed in decibels (dB) with respect to the gain at 1000 hertz for any other frequency. The gain at 1000 hertz shall be taken as the zero dB reference level when the amplifier is adjusted to produce rated power output across its rated noninductive load.

6.2.4 Signal plus noise-to-noise ratio. - The signal plus noise-to-noise ratio is the ratio of the signal plus noise to error at rated power output to the noise level when the signal is removed with no change in operating controls. It shall be expressed in dB with the noise level, in the absence of a signal, being taken as the zero dB reference level.

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6.3 Environmental Testing. - Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.4.3.2).

Preparing activity:
Navy-EC

(Project 5820-N452)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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